



Linux OS – Lab 2

Lab Description:

In this lab, you will explore the Linux operating system and gain hands-on experience with fundamental tasks. You will boot into the CentOS operating system, navigate its interfaces, and perform common operations using the terminal. This lab complements the theoretical concepts covered in the lecture.

Task 1: The Role of the Root User in Linux

- What is the purpose/responsibilities of the root user in Linux?
- What are the risks of using the root user account?

Task 2: Exploring the Linux File System

- Explore the various directories in the CentOS Stream 9 familiarize yourself with their structure and purpose.



Task 3: File and Directory Management

– Working with Environment Variables

- Use the **echo** command to view the contents of the following environment variables: `$PATH`, `$HOME`, `$USER`, `$EDITOR`, `$PS1`, and `$PS2`.
- Display all environment variables in your current shell.
- Customize `$PS1` and `$PS2` to styles of your choice.

– You may use `PS1="\<\t \u@\h \w>\$ "`

– Navigating and Creating Files

- Navigate to `/home/<User>/Downloads` using the **cd** command.
- Create three text files (**file1.txt**, **file2.txt**, **file3.txt**) using the **touch** command.
- Add text to the files and use the following commands to display their contents: **cat**, **tac**, **less**, **head**, and **tail**.
- Display the first 5 lines of the `/etc/passwd` file using **head** command.
- Execute the **wc**, **stat**, and **file** commands on **file3.txt**.

– Directory Operations

- Create two directories: **Data** and **Other**.
- Copy **file1.txt** into the **Data** directory and move another text file (**file4.txt**) into the same directory.
- Delete the **Other** directory using **rmdir**, and then delete the **Data** directory using **rm**.

– Disk Space Usage

- Display the disk space usage of **file1.txt** using the **du -ah** command.